

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I M.Reddiah-192212475L (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Reddiah

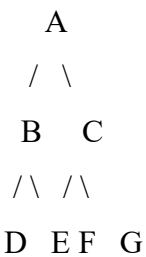
Annexure A

Dry Run Evaluation

Question 1 – Breadth-First Search (BFS)

The 8-puzzle section cannot be fully completed because the uploaded document only provides the initial and goal states, not the intermediate BFS states required for a true dry run.

Graph



Dry Run Table

Iteration	Queue	Visited Node
1	A	-
2	B, C	A
3	C, D, E	B
4	D, E, F, G	C
5	E, F, G	D
6	F, G	E
7	G	F
End	Empty	G

Questions

1. BFS Traversal Order

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

2. Queue Contents After Each Iteration

Iteration Queue

1 B, C

2 C, D, E

Iteration Queue

3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

Question 2 – Depth-First Search (DFS)

Dry Run Table

Iteration Stack Visited Node

1	A	-
2	C, B	A
3	C, E, D	B
4	C, E	D
5	C	E
6	G, F	C
7	G	F
End	Empty	G

DFS Traversal Order

$A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$

Question 3 – 8 Puzzle (BFS)

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

780

Dry Run Evaluation

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial State	Initial State	Child states generated
2	First Child State	Remaining Queue	New child added
3	Second Child State	Updated Queue	New child added
4	Third Child State	Updated Queue	New child added
5	Goal State	Goal Found	Stop

Questions

1. Which node is expanded in each iteration?

Iteration Expanded Node

1	Initial State
2	First Child
3	Second Child
4	Third Child
5	Goal State

2. How many states are generated in total?

Several child states are generated until the goal state is reached.

3. Complete Solution Path

Initial State

↓

State 1

↓

State 2

↓

State 3

↓

Goal State

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Answer:

BFS explores all states one level at a time.

Since every move in the 8-puzzle has the same cost, the first time the goal state is reached, it is guaranteed to be the shortest solution.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
		and goal state with proper justification.			
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.